Advisor Packet (Verbose Version)

Nonlinear Peer Effects in Promotion: Army Evidence and Sports Replication

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I. Executive Summary

This project documents and tests a general principle:

> *Advancement in hierarchical and competitive systems depends not only on individual performance, but also on the **quality of the peer group within which that performance is evaluated**, with **nonlinear effects** arising from the interaction of signal quality and congestion.*

Using U.S. Army officer data, we find an **inverted-U relationship** between peer pool quality and promotion to Major. We propose and implement replications in sports settings—most prominently NCAA basketball—to test whether this pattern generalizes.

II. Core Empirical Finding (Army Setting)

Institutional Structure

- Officers are evaluated within **senior rater × time pools**
- Evaluations include **top-block constraints** (scarce top ratings)
- Promotion is **capacity-constrained** (finite slots)
- Evaluation is **explicitly relative within pools**

Empirical Result

> *Promotion probability increases with pool quality at low–moderate levels, then flattens or declines at the highest levels.*

Mechanism (Careful Interpretation)

1. Signal Quality Effect (Positive)

- Stronger peers → more informative comparison set
- Strong performance in strong pool → higher credibility

2. Congestion Effect (Negative)

- Many strong peers → harder to stand out
- Limited “top signals” and promotion slots
- Reduced marginal promotion probability

Key Insight

Advancement is a function of: - individual performance
- **and the structure of the evaluation environment**

III. Replication Strategy: Why Sports?

Sports settings provide: - Observable performance - Natural peer groups (teams) - Competitive, capacity-constrained advancement - Relative evaluation environments

Goal: > Test whether the **inverted-U relationship** appears outside the military.

IV. Mapping: Army → Sports

Army	Sports
Senior rater pool	Team × season
Top-block rating	Performance metrics / role
Peer quality	Average teammate ability
Promotion	Draft / pro entry / survival
Capacity constraint	Draft slots, roster limits

V. Replication Designs

A. College Basketball → Professional Entry (Primary)

Outcome

- Drafted OR enters professional basketball

Mechanism Mapping

- Moderate-quality teams:
  - high visibility
  - strong signal environment
- Extremely strong teams:
  - congestion (limited touches, role compression)
  - reduced distinctiveness

Why This Is the Best Design

- Clean, discrete advancement event
- Strong analogy to promotion
- Intuitive to reviewers

B. NBA → Career Longevity (Secondary)

Outcome

- Years in league / hazard of exit

Mechanism

- Moderate-quality teams:
  - development and exposure
- High-quality teams:
  - reduced minutes
  - role crowding

Strengths

- Compatible with survival analysis
- Repeated exposure to peer environments

Limitations

- Influenced by injuries, aging, contracts

VI. Fully Specified Empirical Design (College Basketball)

Data Sources

- Sports-Reference (college performance)
- Basketball-Reference (NBA outcomes)

Sample

- NCAA Division I players (2000–2023)
- Player × season observations
- Minimum playing time threshold

Variables

Outcome

$Y_i = 1$ if the player enters professional basketball within k years

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Individual Performance

- BPM (preferred)
- Points per minute
- Efficiency metrics

Pool Definition

Team × season

Pool Quality (Key Variable)

Leave-one-out mean teammate performance

Controls

- Year fixed effects
- Position
- Class year
- Team success
- Conference fixed effects

Empirical Specification

$$\Pr(Y_i = 1) = f\big(\mathrm{Perf}_i,\, \mathrm{PoolQ}_{jt}^{(-i)},\, \big(\mathrm{PoolQ}_{jt}^{(-i)}\big)^2,\, X_i\big),$$

with controls X_i (fixed effects and covariates). Key test: negative coefficient on $\big(\mathrm{PoolQ}_{jt}^{(-i)}\big)^2$ (e.g. $\beta_3 < 0$ in a linear index with quadratic pool term).

Nonparametric Validation

- Bin pool quality
- Plot outcome vs pool quality

VII. Identification and Robustness

Sorting

- Better players → better teams
  Mitigation: controls, robustness

Mechanical Bias

- Fixed via leave-one-out

Support

- Ensure observations in extreme bins

VIII. Literature (Expanded Context)

Peer Effects & Evaluation

- Paulsen (2022): peer effects in NBA transitions
- Kuehn (2023): adjusting for teammate quality

Career Outcomes

- Barnes (2008): NBA longevity
- Zhu (2025): survival models

Context Dependence

- Oberhofer & Schwinner (2017): peer effects in salary

IX. Contribution

If replicated:

> *Nonlinear peer effects are a **general feature of evaluation systems with capacity constraints***

Implication: - Advancement systems are shaped by **interaction of ability and context** - Not purely meritocratic, not purely contextual

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X. Framing for Discussion

This project contributes to:

- Management Science: evaluation systems and organizational design
- OR / Systems: allocation under congestion
- Business Dynamics: nonlinear outcomes from structural constraints

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XI. Key Questions for Advisor Discussion

1. Outcome definition (draft vs broader pro entry)

2. Performance metric choice
3. Sorting and identification strategy
4. Framing: replication vs theory contribution

XII. Bottom Line

> *Being surrounded by stronger peers helps—until it doesn't.*

The goal of this project is to show: - where that turning point is - and why it emerges across domains